

ARNAV KUMAR SHUKLA

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EDUCATION

Indian Institute of Technology Bombay
Master of Science – Statistics — CPI: 8.68 / 10

Mumbai, India
2025 – Present

Banaras Hindu University (BHU)
Bachelor of Science – Statistics — CGPA: 8.13 / 10

Varanasi, India
2022 – 2025

WORK EXPERIENCE

AI Engineer Intern | *Smart Analytica*

(May'26 – Jul'26)

Intelligent Diagnostic Test Selection

- Authored **60+** diagnostic tests across **Hive, Impala, and Spark** covering slowness and common failure signatures
- Embedded issue descriptions with **nomic-embed-text-v1.5** and ranked **60+** tests via **cosine similarity** against **300+ pgSQL** descriptions, with a engine-specific fallback defaults and **adaptive expansion rounds when needed**

Automated RCA Workflow

- Co-designed **Exception/Slowness** tracks, switching from **diagnostic tests** to **error logs** for the exception cases
- Integrated **KB matching, custom LLM prompts, and persistence**, with fallback to KB matches on call failure

KEY PROJECTS

RAG-QA with Reranking | *Self Project*

 *(Feb'25 – May'25)*

- Built a RAG pipeline of PDFs, chunking & embedding text via **Sentence Transformers**, indexed in **ChromaDB**
- Improved retrieval relevance with **2-stage retrieval**, fetching top-20 from ChromaDB & reranking to select the top-5
- Generated context-grounded answers with **GPT-4o-mini** through LangChain, deploying the pipeline via **Streamlit**

Fraudulent Transaction Detection | *Self Project*

 *(Sept'24 – Dec'24)*

- Preprocessed **284K+** credit card transactions using **PCA** & **StandardScaler** with a stratified 80/20 train-test split
- Addressed class imbalance by comparing **SMOTE** & **RandomUnderSampler**, tuning for precision-recall tradeoff
- Fitted **LogisticRegression, RandomForest** & **XGBoost**, achieving **0.84 Recall, 0.83 F1-score, 0.86 PR-AUC**
- Applied **K-Means** for unsupervised fraud pattern discovery, achieving a **0.83 Silhouette Score** on the testing data

Named Entity Recognition | *Self Project*

 *(Jul'25 – Sept'25)*

- Automated **entity extraction** using **TensorFlow**, trained on **GMB** corpus of 48K sentences with an 80/10/10 split
- Captured context using **Embedding, BiLSTM, & TimeDistributed Dense**, achieving **96.2%** masked accuracy
- Built an end-to-end **NER inference pipeline** for text preprocessing, padding, prediction, and entity label decoding

Agentic Stock Market Research Assistant | *Self Project*

 *(Mar'26 – May'26)*

- Designed a **LangGraph** pipeline with 6 specialised nodes: ticker resolution, market data via **yFinance**, news via **Tavily**, sentiment analysis, signal generation, & report writing for automated stock analysis and investment decisions
- Implemented **adaptive retry logic** that re-queries news on low-confidence sentiment before recommendation engine
- Generated BUY/HOLD/SELL signals using **GPT-4o-mini** with **Pydantic v2**, built a **Streamlit** frontend reports

Encoder-Decoder NMT with Bahdanau Attention | *Self Project*

 *(Mar'26 – May'26)*

- Implemented **Seq2Seq Encoder-Decoder** for English to French translation using **LSTM** based encoder & decoder
- Applied **Teacher Forcing** during training on **100K+** English-French sentence pairs & built separate inference models
- Trained two variants with and without **Bahdanau Attention** to compare the role of attention in the translation

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python | R | SQL | C **Libraries:** PyTorch | TensorFlow | Scikit-Learn | Numpy | Pandas | XGBoost
Interests: Generative AI | Agentic AI | NLP **Frameworks:** LangChain | LangGraph | FastAPI | ChromaDB | FAISS
Key Courses: Algorithms | Regression | Time Series **Ongoing Courses:** Deep Learning | Multivariate Analysis
Certifications: Machine Learning Specialization (Coursera) | DS & ML Program (Edureka) | DL Bootcamp (IITB)

POSITION OF RESPONSIBILITY

Interview Coordinator | *Institute Placement Team, IIT Bombay*

(Oct'25 – Dec'25)

- Facilitated placement of **2300+** students collaborating with a team of **250+** members across all departments
- Coordinated assessment tests for leading firms & managed end-to-end placement logistics efficiently throughout drives

ACHIEVEMENTS & EXTRA - CURRICULAR ACTIVITIES

- Secured **All India Rank 33** among **2880 students** in the **IIT JAM Mathematical Statistics Examination**
- Cleared the screening round of **Convolve 4.0**, a pan-IIT AI/ML Hackathon jointly organized by 9 IITs (Jan–Feb'26)
- Secured **1st Runner-Up** position in the Intra Hostel GC Badminton Tournament at Tansa House, IIT Bombay
- Participated in Squash & Badminton Intra-Department Competitions in Mathematics Department at IIT Bombay

HOBBIES

Fitness | Swimming | Table Tennis | Carrom | Badminton | Trekking | Reading Comics | Listening Music | Travelling